

Exam Notes:

- Use the attached bubble sheet to answer the following questions.
- No more than one bubble should be filled in per problem.
- Nodes of equal evaluation in search trees are expanded in alphabetical order.

Chose only ONE answer for each of the following questions

1	 is an <u>uniformed</u> search algorithms	a) A* ☒	c) MaxMin
2	Time and space complexity of <u>Depth First Search</u> (DFS) are.....	b) Greedy ☒ a) $O(b^m)$ and $O(b^m)$ ☒ b) $O(b^m)$ and $O(bm)$ ☒	d) BFS ☒ c) $O(b^d)$ and $O(b^d)$ ☒ d) None of the above
3	Inferencing using Truth Table method is	a) sound but incomplete ☒ b) not sound but complete c) sound and complete ☒ d) neither sound nor complete	
4	the theory proving is an example of an AI agent that: <u>FIFO</u> <u>priority</u>	a) Thinks rationally ☒ b) Thinks humanly	c) Acts rationally d) Acts humanly
5	In <u>Breadth First Search</u> (A*) the frontier is implemented as	a) FIFO queue b) Priority queue ☒	c) LIFO queue d) None of the above
6	It is possible to design a Linear regression algorithm using a neural network?	a) true ☒ b) false	
7	Suppose you are working on weather prediction, and your weather station makes one of <u>three</u> predictions for each day's weather: Sunny, Cloudy or Rainy. Would you treat this	a) regression problem b) classification problem ☒ c) regression and classification d) none of them	
8	A* algorithm (<u>graph version</u>) is guaranteed to get optimal solution provided that	a) Heuristic function is admissible b) Heuristic function is consistent ☒ c) It is always optimal d) No guarantee for optimality	
9	In a CNN model, the input image <u>28*28</u> and the kernel size <u>5*5</u> with stride 1 then apply max pooling <u>2*2</u> . what is the size of the final fracture map generated by these layers <u>28-4</u> <u>24/2</u>	a) 5 * 5 b) 26 * 26	c) 13 * 13 d) No answer ☒
10	what is the number of parameter is required for number of filter <u>20</u> with size <u>5*5</u> in CNN if the input image is <u>28*28</u> <u>20 * 5 * 5</u>	a) $5*5*20*28*28$ b) $5*5*20*26*26$	c) $5*5*26*26$ d) $5*5*20$ ☒

- 11 In unsupervised learning Algorithms
- a) inputs and outputs are required ✗
 - b) only inputs are required ✓
 - c) not required inputs
 - d) inputs are random values

Using the following confusion matrix output form image classification model to answer the following question (from 12 to 15)

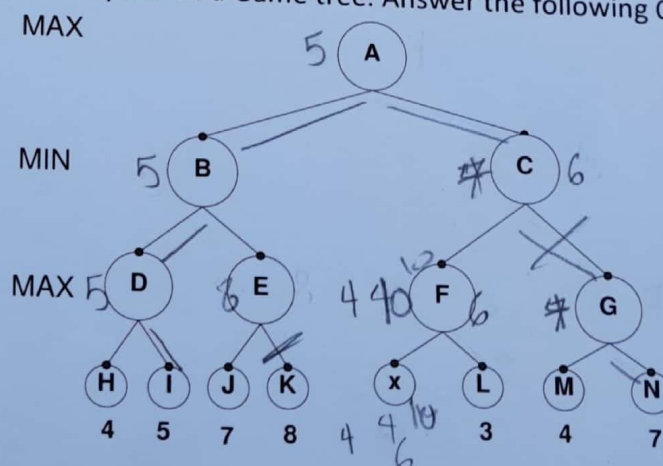
		- Predicate +		
		Cat	Dog	Lion
Actual	- Cat	45	2	3
	+ Dog	2	57	1
	+ Lion	1	2	37

50
60
40
37
37+4

- 12 What is the Accuracy of the model
- a) 92.56%
 - b) 93.89%
 - c) 95.82%
 - d) 89.52%
- 13 What is the Precision for Lion classe
- a) 92.5%
 - b) 20.24%
 - c) 84.09%
 - d) 90.15%
- 14 what is the TN for the Lion class
- a) 3
 - b) 4
 - c) 37
 - d) 106
- 15 what is the FP for the CAT class
- a) 3
 - b) 5
 - c) 97
 - d) 94
- 16 You are given two consistent heuristics that estimate the cost between some node A to the goal G, if $h_1(A) = 10$, and $h_2(A) = 7$. we are saying that.....
- a) h_1 is more informed than h_2
 - b) Both are equally informed
 - c) h_2 is more informed than h_1
 - d) Cannot determine
- 17 In search problem formulation, Which of the following is taking state and action then return the new state?
- a) heuristic function
 - b) transition model
 - c) Available Actions
 - d) None of the above.
- 18 The evaluation function $f(n)$ used in GBFS and A* algorithms respectively are
- a) $h(n)$ and $g(n)$
 - b) $g(n)$ and $h(n)$
 - c) $h(n)$ and $g(n)+h(n)$
 - d) None of the above

Answer questions from 19 to 22

Consider the following tree represent a Game tree. Answer the following Question



- 19 which value of x that make minimax algorithm will select the path $A \rightarrow C \rightarrow F \rightarrow L$
- a) 1
 - b) 6
 - c) 8
 - d) cannot reach this pa



20	what is the path will be obtained using Minimax Algorithm if $x = 10$	a) $A \rightarrow C \rightarrow G \rightarrow N$ b) $A \rightarrow B \rightarrow D \rightarrow I$	c) $A \rightarrow B \rightarrow E \rightarrow K$ d) $a \rightarrow C \rightarrow F \rightarrow X$
21	which value of x that make α - β pruning algorithm Not prune G	a) 3 b) 4	c) 5 d) 6
22	if $x = 4$, Applying α - β pruning algorithm, the pruned branches is accorded at	a) F only b) G only	c) F, and K d) K and G
23	If the target output is Y and estimated output from Neural network is <u>A</u> . The entropy cost function is equal to	a) $Y \ln(A) + (1 - Y) \ln(1 - A)$ b) $Y \ln(A)$ c) $-Y \ln(A) - (1 - Y) \ln(1 - A)$ d) $(1 - Y) \ln(A) + Y \ln(1 - A)$	
24	what is happen if learning rate in linear regression is very large	a) as learning rate increases, cost always decreases b) algorithm may fail to get correct parameters c) algorithm will get in a local minimum d) as learning rate increases, algorithm will always get the correct parameters	
25	if the NN is perform <u>bad</u> in train and <u>test</u> samples what is the name of this problem	a) over fitting b) under fitting	c) bad fitting d) good fitting

Answer questions from 26 to 32

Consider the following problem, predict the length of baby given his/her age and want to find the relation between them, We collect 4 sample as given in the following table

x1: age (month)	y: length(cm)
2	0.58
5	0.66
10	0.73
15	0.77

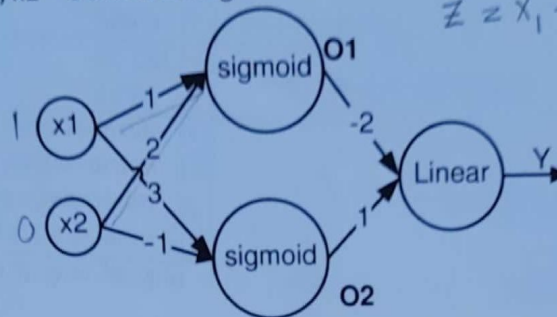
Apply the stable learning algorithm for this problem for 2 iteration to update the parameter of the function $h_{\theta}(x) = \theta_0 + \theta_1(\text{age}) + \theta_2(\text{age})^2$
given learning rate = 0.0001, initial parameter $\theta_0 = 0.1, \theta_1 = 0.1, \theta_2 = 0.1$

26	What is the type of this problem	a) supervised regression b) supervised classification c) unsupervised regression d) unsupervised classification	
27	what is the total cost value given the initial value	a) 58.52 b) 82.22	c) 23.25 d) No answer
28	what is the new value of θ_0 after applying the first iteration (round to 3 decimal point)	a) 0.001 b) 0.199	c) 0.099 d) No answer
29	what is the new value of θ_1 after applying the first iteration (round to 3 decimal point)	a) 0.088 b) 0.188	c) 0.199 d) No answer
30	what is the new value of θ_2 after applying the first iteration (round to 3 decimal point)	a) 0.159 b) -0.059	c) 1.05 d) No answer
31	A sentence α is satisfiable if and only if	a) $\neg \alpha$ is valid b) α is true in all models c) α is true in some models d) both (a) and (c)	

32	what search algorithm guaranteed to get the optimal path to the goal	a) A* b) BFS	c) GBfs d) UCS
33	in CNN models the max pooling layer ...	a) share the weights on the edges b) reduce the complexity of CNN c) generate the same size of the input x d) maximize the cost function x	

Answer questions from 34 to 41

Given the following Neural network. Do the back propagation algorithm for one iteration to make $Y = 1$ for the input $x_1 = 1$, $x_2 = 0$ and learning rate $= 1$



34	what best describe this Neural network	a) 2 input, 1 output and 1 neuron in hidden layer b) 2 input, 2 output and 1 neuron in hidden layer c) 2 input, 2 output and 2 neuron in hidden layer d) 2 input, 1 output and 2 neuron in hidden layer	
35	What is the equation of O1 ..	a) $O1 = x_1 + 2 \cdot x_2$ b) $O1 = \frac{1}{1 + e^{x_1 + 2 \cdot x_2}}$ c) $O1 = e^{x_1 + 2 \cdot x_2}$ d) $O1 = \frac{1}{1 + e^{-x_1 - 2 \cdot x_2}}$	
36	what is the value of O1, and O2 for the given input ($x_1 = 1$, $x_2 = 0$)	a) [0.73; 0.95] b) [1, 3] c) [3, 1] d) None of the above	
37	what is the value of Y for the given input ($x_1 = 1$, $x_2 = 0$)	a) 0.92 b) -0.92 c) -0.51 d) None of the above	
38	that is the absolute error if target output is 1 for the given input ($x_1 = 1$, $x_2 = 0$)	a) 0.92 b) 1.25	c) 1.51 d) 0
39	what is the error at O1 and O2	a) [-3.02 ; 1.51] b) [2.41 ; 4.82] c) [1 ; 0] d) None of the above	
40	what are the new parameter $\theta^{(2)}$ values after apply propagation algorithm	a) [-1.76, -2.29] b) [-0.9, 2.43] ✓	c) [0.5, -1.29] d) [0.73, 0.95]
41	what are the new parameter $\theta^{(1)}$ values after apply propagation algorithm	a) [0.4, 2; 3.07, -1] b) [1, 3; 0.05, 2.89] c) [2, 0.05; 2.89, 1] d) [-1, 2.89; 3, 0.05]	

42	Give the number of rows in the truth table for the compound statement. $(p \vee q) \wedge (r \vee s) \vee t$ 5 25	a) 10 b) 25 c) 8 d) 32
43	the statements are equivalent. $\neg (q \rightarrow p)$ 79VP 9A7P	a) $q \wedge \neg p$ b) $\neg q \wedge \neg p$ c) $q \vee \neg p$ d) $\neg q \vee p$
44	What is the logically equivalent to the following statements "I pass if you pass" Not that fail is equivalent to not pass you pass $\rightarrow$ I pass	a) You pass only if I pass x b) If you fail then I fail x c) If you pass then I pass x d) You fail if I pass
Answer questions from 45 to 49 consider we what to use logistic regression to build AND gate between two inputs, using the following equation $h\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$ and threshold is 0.5 out = 1 if $h\theta(x) \geq 0.5$ out = 0 otherwise given learning rate = 1, initial parameter $\theta_0 = -1, \theta_1 = 1, \theta_2 = 1$		
45	what is the value $h\theta(x)$ at $x_1 = 0$ and $x_2 = 0$ (for given initial value)	a) 0.5 b) 0.73 c) 0.27 d) 0.35
46	what is the output after threshold at $x_1 = 1$ and $x_2 = 0$ (for given initial value)	a) 0 b) 0.5 c) 0.73 d) 1
47	what is the new value of θ_0 after applying the first iteration (round to 2 decimal point)	a) -1.25 b) 0.75 c) -0.75 d) -1.50
48	what is the new value of θ_1 after applying the first iteration (round to 2 decimal point)	a) 1.16 b) 1.25 c) 0.94 d) 0.50
49	After the first iteration, the new parameters of the equations fit the Gate	a) AND b) OR c) XOR d) NAND
50	Having the two rules; $(A \Rightarrow B)$ and $(\neg B)$ we can conclude the rule saying	a) A b) $\neg A$ c) B d) $A \wedge B$

$$\neg A \vee B \wedge \neg B$$

$$\neg A$$

مع أطيب التمنيات بالنجاح

	x_2	x_1	y	z	h	error	error x_1
1	0	0	0	-1	0.27	0.27	0
1	0	1	0	0	1	1	1
0	1	0	0	0	1	1	0
1	1	1	1	1	0.73	-0.27	-0.27
						2	0.73
						0.32	0.18